ActInf Textbook Group ~ Cohort 1 ~ Meeting 4 (Chapter 2 pt. 1)

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Hello everyone, it's May 26th, 2022. It's the fourth week of the first cohort of the
ACTIM textbook group. We're in our first discussion of chapter two. Today we're going
to go over the questions that are posted for chapter two. Then we'll see where that takes
us and what ideas of the book have been explored and that'll be the main focus. Also, just a note
on the math learning group. Those meetings have been occurring on 19 UTC on Wednesdays and so in
the math learning group you'll find information about how to participate. You'll find and be able
to edit resources on learning different topics. So this is a really helpful activity if people want
to share the resources that help them learn about different ideas and that'll help people search for
a video about this. We have a notation table that will help us as we start to look at the notation
more. There's some math learning basic questions like around math. Anything related to the textbook
content can go in the main questions. Math learning group is part of the real textbook group. So these
are not the questions to be answered like the other ones that we're going to go to are, but this is just
about math. And then also there's some math oriented overviews like I wrote this summary of the first
two-thirds of chapter two. So if people want to be like continuing to distill and represent what
they're learning and what they're asking, that's all good. Okay. Any other notes that anybody wants to
add before we start to go into the questions? You can raise your hand or you can type it in the chat.
Okay. So we're in the first week of discussing chapter two, the low road to active inference.
So just to kind of warm us up, what is the low road?
Okay.
Yeah. The low road is the Bayesian brain approach right here.
It's the like proximate next, like big concept, I guess, previous.
But it's kind of based on, I guess, yeah, I mean, you'd see right in the picture, like it's based on Bayesian.
It's like mathematical. The way I conceptualize it is like literally, you know, the low road, meaning like
very primitive, very mathematical, very like hyper analytical, foundational kind of
an analysis way of looking at it.
I guess that's not necessarily living or anything like that.

Whereas the high road is kind of more about the other side of that.
Okay. So we're going to be learning about the high road and the low road.
Those are the two paths that are offered in chapter two and three of the book.
So hopefully again, like it would be awesome to have as many people as possible speak and feel welcome to share their views during this session.
So everyone is especially encouraged to put questions or comments in the chat or raise their hand to speak.
Or find other times to have synchronous communications and especially to make contributions to the coda.
But really appreciate that everyone has come out to go through these discussion questions.

So we're going to now go to the questions that have been placed into the questions table.

And just the more questions, the better.

They can be super short.

It can be anything that you're having uncertainty about.

If you come across a table or a figure or a sentence in the text and you have a question, please write the question down.

It will be extremely, extremely helpful.

It doesn't have to be with extensive references or citations.

It can just be, what did this mean?

If you're uncertain about something.

And then if you were certain about it, you could still write a question like, what would you ask somebody to test or to understand what they understood about it?

Okay. So we're going to start with this first most upvoted question.

But there's still time in this session and also till next week in our next session on Chapter 2 to add more questions if you wrote them down somewhere else and also to upvote questions.

But we're just going to start with the most interestingly voted questions.

So what does anyone see in table 2.1 and what are support and surprise?

Mike, yes.

Mike, yes.

So table 2.1 looks to me like a guide for selecting distributions.

If you're building out one of these models with support saying, you know, what set of numbers would fit within this distribution selection.

And then if you chose this distribution, how would you look at the support equation for it?

Can you unpack what you mentioned with a surprise part?

Yeah. So if you chose this distribution, how would you look at the surprise component of it?

And we're going to get to surprise and Bayesian surprise in the coming questions.

So support using the definition pulled here from Wikipedia.

The support of the function is the subset of the x-axis like the domain where there's a y value that's not zero.

So it's where the function is non-zero defined.

The Gaussian, the normal Belker of distribution, has non-zero support at all numbers.

Fancy R.

And then other distributions have different variable, different values that they're having non-zero values at.

Like the gamma distribution is defined over the interval from zero to infinity.

So if it was like, how many of this thing do you have?

That might be a reason to use a distribution that's only positive numbers.

Or if you were wanting something that could have positive or negative numbers of any size, the Gaussian is a distribution.

And then another one that's like a relevant support difference would be like the Dirschle being defined only within the zero to one interval.

So, which helps connect it to probabilities.

So the support is where that function, that distribution family is defined.

And then what is the surprise?

Does anyone know the name of the little squiggle they are using to represent surprise?

I think it's called Fracture Eye.

I.

The first, we're going to come to the definition of surprise in a coming question.

But, and without going into every variable and trying to overlearn something, the surprise of a given observation, like a data point, let's just think about the Gaussian.

X is the observed data point.

And mu is like the mean of that normal distribution, which is also the mode.

So it's like the sensor of that distribution.

And then this capital pi is a variance estimator, I believe here.

So how surprised one should be by a new data point coming in is related to the difference between that new data point and the mean.

So if the new data point were exactly on the center of the distribution, this whole term is going to go to zero.

So one should not be surprised at all by exact data points on the mean of the distribution.

But any data point that has not exactly on the mean is going to have some non-zero value.

And so it's going to be a function of how far the data point is from the mean and some scaling by the variance.

So if we were expecting 100 with a variance of 10 versus 100 with a variance of 1.

In one case, like getting 101 is like a 1 sigma event.

It has a z-score of 1.

Just speaking loosely here.

Whereas in the case of it having a variance that's very small, then that might be more surprising.

Any other comments that people have on table 2.1?

I have a question.

Yeah.

So while I was reading through the chapter,

going back to section 2.2, right?

Perception as inference.

So when they say that the brain is a protective machine or a statistical organ that infers and predicts external states of the world.

Okay.

Here's my question.

Because I'm primarily from a telecom background.

So most of our things are to get messages crossed between two distant transmitters and receiver, right?

So we build noise models.

So without any noise, you don't get signal.

So you need to know what a noise model looks like.

And then, you know, so for example, if the power spectrum of the signal is what you expect from, let's say you're using a Gaussian distribution of noise, right?

So if it's the same as you expect from the Gaussian, you do nothing.

So that's just noise.

So you don't classify the data that's coming in.

If it is more than this, then okay.

So there might be something interesting about this.

There might be some signal here.

So we define the signal to noise ratio.

A lot of this chapter seems to be dealing with the fact that we by default know what the signal looks like.

So there's no attempt to build a noise model here.

There would be like, okay, so maybe the sensor in this case, our eyes, for example, or has some parameters, right?

It can detect between, say, a bandwidth, well, okay, so it can detect a light between certain frequency ranges.

But maybe it performs very well in a certain frequency range and not others.

And there's more error in other frequency ranges, right?

So the noise model should incorporate such things.

Imperfections in the sensor.

Imperfections in the connections between the sensor and processing equipment.

But there's no attempt to do that in this entire chapter.

So I'm just curious as to what exactly is the predictive machine here?

What is being predicted here?

Great question.

And in general, every question that people have, they're going to be 100 times more impactful if they can be written down.

This is just the second chapter.

It's introducing us to the essence of starting with just Bayes' equation.

It's the first equation here.

Starting with the foundation of increasingly advanced noise filtering approaches.

So check out Livestream 43 to see how Kalman Filter and advanced multi-level Bayesian filtering schemes are used and learned.

And fit with free energy minimization and so on.

But you're right that it is not approached in this chapter.

Because it's starting with the essence of Bayes' equation.

And then going to build towards another kernel, which is going to be the kernel of the variational free energy.

And then that, in this one layer case, is going to be able to be elaborated and parameterized and fit within nested models to accommodate learning on hyperparameters, learning of noise models, and so on.

But it is not in this chapter.

Okay.

Yeah, so I was just, I'm sorry if I threw us off track.

No, so it's perfect.

But they're good questions.

And then Jakob asked, what do you think of the surprise of the distributions involving a sum product of the product probability distributions except for gamma?

Actually, I guess the Gaussian kind of as well.

Yeah, maybe I'm thinking about this wrong.

But I thought that the surprise was like the, I guess, the information gain or equivalent.

And so for the Gaussian and the gamma distributions, it's always for one single element of the distribution for one X.

But for the multinomial and Dirichlet, there's something over all of the possible probabilities, which makes it, I guess, an entropy function.

But is there a difference in the distribution?

For example, one could have a multidimensional Gaussian, a two-dimensional Gaussian.

And then the surprise for the two-dimensional Gaussian would be represented as a sum over the different dimensions.

So they've shown the Gaussian and the gamma in the unidimensional case and the multinomial and the Dirichlet in a multidimensional case.

So in that case, is it, I thought that X was just one element of the domain.

So that's not necessary.

So X is just, if we think of the Gaussian as this kind of bell-shaped curve on the X and Y axis, we really want the F of X value to get like the Gaussian value.

But X is just the input value.

Is that correct?

I believe that is accurate because, yes, X is the domain of the function, and then the sum is over i's.

So if we had a multidimensional Gaussian in i dimensions, then it would have a sum over i's.

Yeah, so if, but wouldn't, I guess I still have uncertainty about why we don't do that for the single, single dimensional Gaussian as well.

Because if we want to, if we take just one point on the Gaussian, I understand why that would have a certain surprise.

But then in the same way, if we take one point in the Dirichlet distribution, wouldn't that also, shouldn't that also have an equivalent equation for surprise?

Why in the Dirichlet distribution are we summing over all the i's, but in the Gaussian we are considering just one point on the probability distribution?

Or maybe Dirichlet is multi-dimensional and that's the reason.

I think this, the unidimensional case would be just without this sigma i.

It would just be this.

But if anyone has any other thoughts, especially if they want to contribute them in, like, writing to improve our understanding, that'd be awesome.

Let's continue to the next question.

In section 2.4, they write,

We now introduce the simple but fundamental advance offered by active inference.

This starts from the same inferential perspective discussed above, but extends it to consider action as inference.

In your own words, what is the simple but fundamental advance offered by active inference?

So what would anyone write or like to share?

What is the advance of active inference?

Is it just bringing action into the equation?

So,

predictive coding and Bayesian brain are specifically about perception,

visual perception.

I think the key thing with active inference is that it's,

well, it's not just that it brings action into the equation,

it's that it treats them as essentially the same thing that,

that kind of united

as part of the same process,

which I think is,

I think is kind of a new development.

Thanks for the awesome answer.

Ali?

Yeah, I think Alan Bartho,

or Bartho's in The Brain's Sense of Movement,

has a particularly

relevant

description of action and perception,

and he writes that perception

is simulated action,

and I think that

really speaks to

the advance made by active inference,

I mean, the integration of action and perception,

or, in other words, as Bartho claims,

the simulation of action and perception

is, I guess, a really novel development.

I think I read something similar.

But I can't remember where I read it,
but I think it was roughly that
on these existing Bayesian brain theories,
an agent perceives in order to act,
whereas under active inference,
an agent acts, action kind of precedes perception.
So active inference essentially flips this traditional schema
in terms of which one feeds into the other.
Yeah.
I was just going to say that, yeah,
adding on to bringing action into the equation,
it flips it.
Like, there's no...
Like, where do your priors come from?
Where do the hyperparameter get specified?
Like, where does the model start?
Where is the agency, kind of,
in all of these other models?
It's really no affordance for that.
Or it's hard to...
It's, like, very emergent.
Whereas here, it's, like,
very explicitly specified
some affordance for that.
Thanks, Brock.
Mike?
And then anyone else who raises their hand.
Yeah.
So I'm thinking,
if you have priors,
you have some prior assumptions,
then do you necessarily need to take action
in order to sort of envision
what your perception might be?
So it's almost like getting into imagination
or something like that.
You have a set of priors
that can drive

what you imagine your perception to be
before any action is taken.
Yeah.
Great.
Question?
Blue?
So I think
that this was, like,
elaborated on a lot
in Section 2.7
when they talk about
pursuing a policy.
Because there's consequences.
Like, when we're planning
and making predictions,
that has consequences
for both our perception
and our action.
Like, when,
I mean, even just planning,
even in a conversation,
when someone says,
oh, have a nice, like,
have a nice day
is expected at the end of that.
But, like, have a nice rabbit
would be like, what?
Like, you're not expecting
that to come
at the end of that sentence.
So when you plan,
it interferes with your perception
because you didn't plan for them
to say have a nice rabbit, right?
But it also interferes
with your action.
And so I think, like, this,
the, like, novelty

or enhanced aspect
of active inference
loops into the fact
that perception and action
are continually impacting
one another.
Yes.
Thanks, Blue.
Um,
these are some of the core
memes and themes,
and it really is important
to understand
how active inference
is similar and different
than other work
in this area.
like,
variational Bayesian inference
is not introduced
by active inference.
Bayesian models of perception
is not introduced
by active inference.
Bayesian models of action
is not introduced
by active inference.
So,
it's about
finding what has been done
to understand
what is being offered
and then whether one
chooses to take this, like,
history of science,
development of science
view,
or we kind of just want to

state plainly what it is
without wondering
what the advance
relative to other frameworks is.
These are all really
critical ideas,
like,
the signal processing framework
was brought up earlier,
and predictive processing,
predictive coding,
anticipatory systems,
are often
purely
about sense.
And,
just to give one thought
on that,
it kind of makes sense
for a video encoding
algorithm,
for example,
because every piece
of the camera
is in focus.
But,
vision
requires action,
and so,
that's where
we're going to be
bringing in all these
other important concepts,
like,
attention,
sensory
attenuation,
and

the active
decision-making
components
of vision
to resolve
uncertainty,
which is what
gives rise
to a generative
visual field
that seems like
there's color
everywhere,
and seems like
there's high-resolution
everywhere,
though that is not
the incoming
sensory information.
So,
by fully taking
this generative
modeling perspective
on perception,
which,
in livestream
number 43.0,
Maria did an awesome
job
of connecting
this
even before
Helmholtz and
Kant.
It's Plato's cave,
and it's part of
this long
discussion about

perception,
and action
could be a
variable.
So,
there's perceiving,
and then if action
is a variable,
then active inference
is providing
not just a unified
framework,
like a Bayesian
graph framework,
for some variables
that are interpreted
as perception,
and some variables
that can be interpreted
as action selection,
or policy planning,
but
there's a tractable
approximation
by shifting from the,
and bringing this
answer,
by shifting from
an exact solution
to the mathematical
problem,
Bayesian inference,
to an approximate
solution,
variational free energy,
which is going to
provide a bound
on some

quantity
that might be
intractable to compute.
So,
it's going to give
a heuristic
or the action
perception cycle.
But this is a really
great question,
so people can
continue to return
to it,
because people will
probably ask us
for a long time
to come,
how is this
different than
blank,
or what is
active inference?
these are some
of the things
that we would
want to have
in mind.
Ali?
Yeah,
again,
going back
to Barthaud
and his definition
of perception
as simulated action,
I think it
also relates
nicely to

Merlepanti's
phenomenology.
Merlepanti
has a famous
statement,
and has a famous
insight as
vision is the
brain's
way of touching.
And so,
you see,
for instance,
as we move
around space,
as we
construct our
sense of
spatiality,
I mean,
or our
sense of
temporality,
or everything
else,
we don't
just begin
with the
representation
of space
or time.
I mean,
the image
of movement
constructs
because we
move,
and not

as a
consequence
of our
movement.
Thanks.
And we'll
continue with
the questions,
but that's
an awesome
area to go
into with
phenomenology,
with 4E
cognition,
extended,
embedded,
in culture,
et cetera,
et cetera.
And many,
many of the
live streams
and papers
have been
on those
areas.
So it's
cool to
bridge from
formal models
of perception,
cognition,
and action
into
qualitative
and
philosophical

areas.
Okay.
And then,
oh,
thanks,
Moritz,
for adding
that.
The idea
of interrelated
action and
perception
has been
around in
cognitive
science
for a
long time.
Like,
dynamical
systems
theory
and
inactivism
emphasize
that,
but active
inference
brings
a coherent
formalism
to the
table,
which is
a nice
advance.
Well said.
Next question.

Figure 2.2
illustrates
y as a
result of
an internal
generative
model,
x,
and an
external
generative
process,
x star.
y,
in order to
measure
surprise,
wouldn't we
need another
value of
y,
i.e.
a separate
y that
encodes
prior
beliefs?
If y can
be objectively
measured from
external signals,
is there a third
y that's
considered the
observation?
first,
I'd like to
actually go to

this question
about two
notions of
surprise.
So,
because
they might
come to
bear on this
action perception
loop.
Once we
move from
just
Bayes'
equation
and
calculating
surprise on
parametric
distributions
to
thinking
about
prior
updating
and
cognitive
entities
with a
generative
model
that
constitutes
a prior
with
incoming
information

coming in,
we're going
to start
to see
why this
notion of
surprise
and Bayesian
surprise
are similar
and different.
So,
using the
apple frog
example,
so,
the
unobserved
hidden state,
the latent
state of the
world,
is whether
there is
an apple
or a frog
in a bag,
for example,
or just
in this
person's
area,
and then
what can
be observed
just speaking
coarsely
is it can

jump or
not,
and we
can totally
go down
the rabbit
hole with
what is
truly observed
and things
like that,
but this
is just
what are
in the
context
of this
model,
the unobserved
state is
the actual
identity
of the
object,
and the
observation
is going
to be
the action
or not.
We take
the opportunity
to unpack
two different
notions of
surprise,
both of
which are

important.
The first
we refer to
simply as
surprise.
It's the
negative log
evidence where
evidence is the
marginal probability
of observations.
So we saw
that first
sense of
surprise with
the fracture
eye in
the previous
table we
looked at.
The second
notion of
surprise is
referred to
as Bayesian
surprise.
surprise.
This is a
measure of
how much
we have
to update
our
beliefs
following
an
observation.
In other

words,
Bayesian
surprise
quantifies
the difference
between a
prior and
a posterior
probability.
What are
the similarities
and differences
between the
two notions
of surprise?
Okay, we'll
see what has
been said
and then
hear what
other people
are thinking.
So
page 20
they wrote
that
Bayesian
surprise
scores the
amount of
belief updating
as opposed
to surprise
which is
simply how
unlikely or
likely that
observation

was.
So
in that
Gaussian
case,
the surprise
is like
about one
data point
coming in
given how
the
distribution
is
parameterized
right now,
how surprising
was that
one data
point?
And then
Bayesian
surprise is
going to be
about how
much that
distribution
is updated
after
processing
that data
point.
Similarities,
they both
depend on
how well
the agent's
generative

model matches
the external
world.
And they're
both measuring
surprise
in the
same
units,
which are
information
theoretic
quantities
of information,
i.e.
Nats or
bits.
Difference is
plain surprise
marginalizes
over all
the model's
degrees of
freedom under
the model's
prior distribution
over its
adjustable
parameters.
That would
be great
for someone
to unpack
what does
it mean
to marginalize
over the
model's

degrees of
freedoms.
And then
Bayesian
surprise
lets the
model choose
the best
set of
parameters
it can
fit the
observed
data
and then
measures
how much
of an
update
that
was.
Mike?
And then
Blue?
And then
Ali?
Yeah,
under the
first
similarity,
I'm wondering
if it's
this dependency
on how well
the agent's
generative model
matches their
perception

of the
external world
as opposed
to matches
the external
world.
agree with
that.
Great
addition.
Blue?
And then
Ali?
Okay.
I think my
hand was just
left over,
left up.
Okay.
Ali?
Well,
I believe
that the
plain surprise
is a kind
of raw
statistical
surprise,
but Bayesian
surprise is a
kind of
surprise and
is a kind
of processed
surprise,
or I'm not
sure if I'm
right in

saying that
plain surprise
can probably
be described
as an
objective
surprise,
as opposed
to Bayesian
surprise,
as a
subjective
surprise.
But as
I said,
I'm not
sure
about the
plausibility
of these
descriptions.
Okay.
Great
comments.
Even
the
surprise
alone
still
depends on
the
parameters
of the
generative
model.
So it
still is
within a

processing
or filtering
frame,
albeit a
fixed one.
So
they talk
about,
like,
let's look
at the
figure where
there's a
graphical
overview
of this
apple
and the
frog.
So,
and this
is kind
of giving
some
graphics
and words
to fill
in this
apple
frog
example.
So
initially,
the person
has this
likelihood
model in
the back

of their
head,
where they
have beliefs
about how
likely apples
are to
jump,
they do it
1% of the
time,
and how
likely frogs
are to
jump,
they do it
81% of the
time.
That's the
likelihood.
likelihood.
That's
about how
observations
depend on
hidden states
of the
world.
Their prior
beliefs are
that there's
a 10%
chance that
the entity
is a
frog.
And these
are mutually

exclusive
options,
there's no
third option
here,
that would
be like
another
category
model,
structure
learning on
the model,
so we're
staying within
this model
for now.
and they
sum to
one because
they're a
probability.
Then there's
an observation,
which is
jumping,
and then
the posterior
reflects the
updated beliefs
about what
the entity
is.
And so
it's like
it's so
much
overwhelmingly

more likely
that frogs
jump,
that seeing
something
jump
updates the
prior from
here to
here.
So in
this case,
one can
calculate the
surprise of
the observation
without doing
any updating
at all.
One could
just stay
fixed in
their prior
belief,
and then
could describe
how surprised
they are
in knots
by a given
observation.
In this
full Bayesian
cycle,
there is an
updating of
the prior to
the posterior,

and then
that updating
can be
described in
terms of
how much
the prior
was updated.
And so
that's like
here,
the model
is updated
to the
observed data.
This is now
the best-fitting
model.
And then
we're calculating
the difference
between these
two distributions.
So
regular
surprise
being zero
means the
data point
was exactly
as you
expected.
Bayesian
surprise
being zero
means the
distribution
was not

updated.
High
surprise
means that
the data
point was
extremely
unpredicted,
it was
extremely
unlikely
whether or
not you
update your
model at
all.
High
Bayesian
surprise
means that
the distribution
was changed
a lot
as a function
of seeing
that happen.
Ali?
Well,
perhaps I
didn't understand
it correctly,
but isn't it
the case that
in the plain
surprise
or more
generally
in the

Bayesian
formulation of
the probability
of events,
the likelihood
and both
the likelihood
and the
priors
are
somehow
inherent
to the
phenomena,
inherent to
the events,
independent of
the observers?
Yeah,
great question.
I hope I'm
not going
off on a
branch here,
but this is
related to
the difference
between
frequentist
and Bayesian
approaches to
statistics.
So
frequentism
does have,
viewed from
the Bayesian
perspective,

frequentism
does have
priors.
They're
uniform
priors,
which are
not
uninformative
priors.
They just
are uniform.
And so
in frequentism,
we come
across like
the maximum
likelihood
solution or
the maximum
likelihood
parameter,
which is
just like,
well,
if all
outcomes
were equally
a priori
likely,
uniform
prior,
then we
would just
need to
evaluate the
likelihood and
find the

model with
the maximum
likelihood
solution.
Bayesian
offers another
degree of
freedom.
It says,
well,
sure,
you could
pick a
uniform
prior that
would give
you the
maximum
likelihood
solution,
or you
might want
to have
a prior
distribution
over that
space,
so that if
something is
twice as
likely a
priori,
and then you
observe less
than 2x
evidence for
it,
you still

might want
the base
factor or
your posterior
to reflect
that one
rather than
just jumping
instantly to a
different maximum
likelihood solution.
So,
yes,
priors and
likelihood are
implicit,
but they're
using a
different ontology
than Bayesian
statistics,
but we're in
the Bayes or
the post-Bayes
area now.
But there's so
many connections
to classical
statistics,
and in SPM,
the textbook,
there is
parametric
classical
statistics,
non-parametric
classical
statistics,

and Bayesian
statistics,
so they're
more similar
than not.
It just is
about seeing
where one of
them is like
a special
case of
another,
or a
generalization
of another.
Okay,
let's see
if we can
do one or
two more
questions during
this session.
Okay,
so let's
return to
the previous
question.
So we're
looking at
figure 2.2,
which is
something we're
going to see
different
representations
of.
This
entity as

generative
model,
world,
or niche
as generative
process,
and then
here,
the hidden
states are
those that
are unobserved
as data.
y is
referring to
data points
that are
observed as
data.
There's a
cognitive
hidden state,
which is
like a
prior in
this generative
model,
and then
there's some
hidden state
x star,
but it could
have been
any letter
or any
shape,
about the
hidden state

in the
world.
The
observation
is going
to,
everything that
happens in
between here
is cognition,
like the
sandwich
model,
like sense,
think,
act,
that type
of model is
just referring
to that
little boomerang.
Data come in,
cognitive
processing,
action selection.
So we're in
that figure.
The question
was,
in order to
measure surprise,
wouldn't we need
another value of
y, i.e. a
separate y,
that includes
prior belief?
Great

question.
Let's just
assume that
there is a
Gaussian
generative
model.
So the
entity has
like two
parameters in
its cognitive
model,
the mean
and the
variance.
Then y
comes in,
and given
the parameterization
of the
generative
model,
all that's
needed is
the data
point to
come in
for the
surprise
to be
calculated.
So another
value is
needed,
but whether
we call it
ABC

XYZ
is just
a
mathematical
abstraction.
So yes,
prior beliefs
are important.
The prior
beliefs that
can be
interpreted as
the
parameterizations
of the
cognitive
model are
required.
If y
can be
objectively
measured from
external
signals,
is there
a third
y that is
considered the
observation?
So one
could imagine
a lot of
real-world
scenarios where
a more
complex model
would be
required.

Two people
looking at
the thermometer
or all
these different
sorts of
situations.
But unless
somebody can
unpack this
a little
more or
explain what
they were
asking about,
then I don't
think a third
y is required.
Mike?
Yeah, that was
my question,
so I can try
and unpack
it.
And so
sticking with
the thermometer
example at
the heart of
that last
question is,
if we have
a thermometer
that's
registering
the temperature,
can we
consider that

as sort of
the true
observation,
that is the
actual y
as measured
by this
instrument,
and then
therefore we
can compare
our sort of
internal y
with the
actual y?
Okay.
So I have
a really
similar and
related question
that's like
two questions
down.
Okay.
But it's in a
different section,
but let me just
place some things
on here,
and it goes back
to what we were
saying about
what exactly
is the hidden
state.
So is the
hidden state
the temperature

and the data
is the
reading from
the thermometer
that, like,
that's like
the data y,
right?
Like, the out
this y in the
middle.
But then, like,
what is how I
feel hot or
cold?
Like, I mean,
I'm perfectly
positioned to be
at 75 all the
time, so, like,
I know if it's
like one degree
too cold or
one degree
too hot, like,
I've got to
turn up the
heat or turn
it down or
whatever.
But, like,
my registration,
my sensory
input,
is not at
75 degrees.
Like, my
sensory input

is I'm
hotter, I'm
cold.
And so,
this is, like,
where, I mean,
we were talking
about this
yesterday in
the math
group, like,
there's this
really fuzzy
line for me,
like, I would
love for someone
to clarify that.
So, I do get
what you mean
about, like,
this in, like,
shouldn't there
be another
why?
Like, there's
the why out
there in the
world, 75
degrees, and
then there's
how I feel
about that
why.
Okay, thanks.
So,
variables are
not, like,
innately

tagged with
being observables
or hidden
states.
It's a
model-specific
framing of
what is going
to be modeled
as generated
data and
what is going
to be modeled
as a
Bayesian
prior.
In
multi-level
Bayesian
modeling,
the priors
themselves are
generated from
a higher or
deeper level
of the, so,
one can be
serving, in
fact, multiple
roles.
This is the
minimal, prior
generated data
expectation
maximization
type single
layer Bayesian
kernel.

So, we're
going to think
about this
temperature
example.
There's a
hidden state
of the
generative
process, which
is going to
be the
temperature of
the world.
Latent
unmodeled.
So, this
isn't even
claiming that
there is such
a thing as
temperature.
This is just
a latent
unobserved
temperature that
is giving
rise to
thermometer
readings, which
might have
different sorts
of noise.
Then, there's
another hidden
state, which
is the
evaluation of

temperature.
And so,
this is just
a schematic
that's, and
also, whether
one is labeled
X or Y
or L
or triangle,
is like a
norm and
a convenience,
but the
letter doesn't
matter itself.
It will be
used mostly
consistently within
the textbook,
but there's
only so many
letters, and
there's not a
lot of
coherence on
notation use.
What else
could be
explored here?
Mike and
then Ali.
So, as I
keep ruminating
on this, which
is probably
more than I
should, it

seems like there
can be infinite
Ys, right?
So, to the
extent that you
are taking
action, which
will change
your perception,
then you are,
as a result,
triggering
potentially new
Ys in the
system, right?
So, you
just have this
infinite potential
for what Y
can be.
This speaks to
the composability
and the
flexibility of
active inference.
So, like, Y
could be one
pixel of
visual input.
It could be a
4K video.
Y could be
smell and a
4K video.
Y could be
LiDAR and
this and that.
So, Y is

just the
generalized
input of
sense.
And then
action might
be related
to this in
a very
direct way.
Like, Y
could be
visual input
and action
could be
your eye
movement.
So, that
case is
going to be
explored a
lot in
the book.
However, it
could also be
you're getting
something in
and then the
actions are
just totally
unrelated.
Maybe they
don't affect
the hidden
state, the
causal process
in the world
at all.

Or maybe
actions enable
different Ys
to enter
the picture.
This is
just the
total essence
kernel.
And then
even trivial
cases require
some more
apparatus.
Ali?
Yeah, I
have a
question.
Is it
correct to
say that
the minimization
of variational
free energy,
aka the
surprise,
is exactly
equivalent to
the maximization
of expectation?
I mean,
is it
the wholly
linear
relationship
between these
two?
Or possibly

we have
some kind
of plateaued
areas between
these two
extremes?
great question.
So this is
something that
we'll come
to next
week in
our discussions
on chapter
two.
So we
have not
even discussed
variational free
energy yet.
Today we
talked about
starting with
a low road
on some of
the atomic
calculations that
are going to
come into
play, like
surprise,
and Bayesian
surprise in
a Bayesian
framework, and
beginning to
partition active
entities and

their action
perception loops
in terms of a
Bayesian graph
that's going to
be amenable to
a flexible
modeling of
perceptive,
cognitive,
active, and
out there in
the world
variables,
variables with
those interpretations.
And then
variational free
energy is going
to come into
play as a
way to
bound
surprise.
So let's
have some
questions and
discourse, and
we'll come to
it next week, but
it's an awesome
question.
Jessica?
Hi, yes, I
have a question,
I guess, related
to this figure
2.2.

Let's say,
like, in
regards to,
like, how we
interpret and
like, what we
observe.
So like, I
understand, like,
you have a
prior that might,
you know, you
anticipating, you
know, something,
like some other
people's behavior or
like how things
should, that might
be different from
what you actually
observe.
But a lot of
times, at least
in terms of, like,
when we're thinking
like human beings,
like, say, like,
an action in the
world, like, how
you interpret it
and has to do a
lot with,
like, your own
like views or
like your own
experiences.
And maybe that
connects to the

priors.
But like, like,
the interpretation
might be very
different.
Like, from
people to
people, the
interpretation
will vary a
lot because
everybody has
like a lot of
like different
experiences.
So those could
be like encoded,
I guess, in the
priors.
But it's not
connected, per se,
to the
prediction part
or maybe it
is.
So I'm trying
to kind of
connect that
idea of
how, like,
our personal
interpretation
based on our
experiences,
which could be
priors,
make us see
the, like,

what's happening
in the world
differently than
other people.
and so that
why will look
very different
for you than
for me, just
because we
have lived
different
experiences.
So it's not
like an actual
why.
Like, it's not
like you cannot
really say,
like, this is
a fact.
And because
it's an
interpretation
at the end
of the day.
Great
question and
points.
Yeah,
understanding
how
the
individual
setup of
a given
entity is
related to

its past
experiences and
how that could
be modeled by
priors is an
important area.
And
some of
those rich
dynamics are
not in this
minimal
nucleus.
But it'll
be awesome to
start to think
about what
does have to
be in the
box here
to give
rise to
those kinds
of dynamics.
So that's
going to
end this
discussion.
Next week,
we will also
be staying
in chapter
two and
taking it
more towards
variational
free energy
and expected

free energy.
So that
should be
fun.
We're going
to, in
this room,
transfer
immediately to
the .tools
meeting.
So if you
want to
join for
Actinflab.tools
meeting, everyone
is welcome and
there's no
prerequisites or
anything like
that.
If you want
to hang
around and
join .tools,
stay in this
exact gather
space.
If you want
to continue
talking with
other people
about the
book or
anything else,
just head up
into one of
the rooms

above.
And if
anyone who
wants to can
just continue
any discussion
that they
want, but
in this space
we're going to
continue now
with tools.
So, thanks
everyone.